

HOFU AB, Japan

WMO: 477880

Lat: 34.034N

Lon: 131.546E

Elev: 5

StdP: 101.26

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.0	-1.0	-8.1	1.9	2.8	-6.9	2.1	3.4	11.0	5.0	9.9	5.0	2.7	10	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	33.0	26.2	31.9	26.0	30.9	25.9	27.3	30.6	26.7	29.9	26.3	29.3	4.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	28.9	26.0	21.4	28.7	25.2	20.4	28.1	86.5	30.9	84.3	30.4	82.2	29.2	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.8	8.4	7.4	DB	-4.3	34.9	1.7	1.2	-5.6	35.7	-6.6	36.4	-7.6	37.1	-8.8	37.9
			WB	-5.3	28.2	1.5	0.8	-6.4	28.8	-7.3	29.3	-8.2	29.8	-9.3	30.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	5.0	5.8	8.9	13.8	18.6	22.4	26.5	27.6	23.9	18.3	12.5	7.2	(6)
		DBStd	8.23	2.53	2.92	3.08	3.07	2.19	1.95	2.03	1.71	2.59	2.91	3.33	3.03	
		HDD10.0	451	157	122	58	6	0	0	0	0	0	12	96		
		HDD18.3	1772	414	351	292	137	23	1	0	0	36	175	344		
		CDD10.0	2618	2	4	24	121	267	371	510	545	418	258	89	10	
		CDD18.3	897	0	0	0	2	32	122	252	287	168	35	1	0	
		CDH23.3	7427	0	0	0	10	143	536	2303	3071	1232	133	0	0	
		CDH26.7	2232	0	0	0	0	12	67	741	1126	281	6	0	0	
(14)	Wind	WSAvg	3.3	3.2	3.3	3.6	3.6	3.5	3.1	3.3	3.4	3.1	3.0	2.9	3.2	(14)
(15)	Precipitation	PrecAvg	1741	82	82	126	144	173	240	280	172	173	96	95	80	(15)
(16)		PrecMax	2448	175	125	240	290	357	451	622	502	368	225	210	154	(16)
(17)		PrecMin	1192	24	27	62	44	53	48	50	35	53	13	28	13	(17)
(18)		PrecStd	315	36	27	49	60	71	105	171	105	82	59	49	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	14.9	17.0	20.0	24.8	28.1	30.0	34.1	34.3	32.0	27.2	22.2	17.2		(19)
(20)		0.4% MCWB	11.9	13.9	13.4	17.4	17.9	24.6	26.5	26.5	25.8	22.2	17.8	13.5		(20)
(21)		2% DB	12.9	14.9	18.0	22.2	26.1	28.2	32.9	33.1	30.2	26.0	21.0	15.9		(21)
(22)		2% MCWB	9.6	11.7	12.2	15.5	17.7	22.7	26.2	26.3	24.9	21.3	17.1	12.3		(22)
(23)		5% DB	11.2	13.1	16.2	21.0	24.9	27.2	31.8	32.2	29.2	24.9	19.9	14.2		(23)
(24)		5% MCWB	7.9	9.6	11.9	15.1	17.8	22.1	26.1	26.1	24.4	20.3	16.1	10.9		(24)
(25)		10% DB	10.1	11.2	15.0	19.8	23.8	26.1	30.2	31.2	28.1	23.8	18.8	13.0		(25)
(26)		10% MCWB	6.9	8.0	11.3	14.6	17.6	21.7	25.7	26.1	23.7	19.2	15.3	9.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	13.0	14.8	15.8	20.1	21.7	26.0	27.8	28.5	27.1	24.0	20.2	15.5		(27)
(28)		0.4% MCDB	14.1	16.3	17.6	21.6	24.0	28.6	31.5	32.0	29.7	25.7	21.4	16.5		(28)
(29)		2% WB	10.6	12.5	14.3	17.9	20.6	24.7	27.1	27.6	26.2	22.7	18.4	13.1		(29)
(30)		2% MCDB	11.8	14.2	16.3	20.2	23.0	26.8	30.6	30.8	28.7	24.9	20.0	14.9		(30)
(31)		5% WB	8.8	10.6	13.1	16.6	19.7	23.8	26.6	27.0	25.6	21.2	16.8	11.6		(31)
(32)		5% MCDB	10.6	12.3	15.5	19.4	22.5	25.7	29.9	30.3	28.1	23.8	18.9	13.7		(32)
(33)		10% WB	7.2	9.0	11.7	15.6	18.8	23.0	26.2	26.6	24.9	19.9	15.6	10.2		(33)
(34)		10% MCDB	9.4	11.0	14.5	18.5	21.9	24.9	29.3	29.6	27.5	22.6	18.0	12.4		(34)
(35)	Mean Daily Temperature Range	MDBR	8.5	8.9	9.6	10.0	9.4	6.8	6.2	6.8	7.5	9.0	9.2	8.8		(35)
(36)		5% DB MCDBR	9.9	10.5	11.4	12.0	12.0	8.6	8.5	8.3	7.7	9.1	9.3	9.8		(36)
(37)		5% WB MCWBR	7.1	7.6	7.4	6.8	6.2	4.5	3.8	3.6	3.9	5.3	6.1	7.1		(37)
(38)		5% WB MCDBR	8.5	8.7	8.8	8.4	7.8	5.7	7.1	6.8	6.5	7.4	7.8	8.5		(38)
(39)	Clear-Sky Solar Irradiance	taub	0.438	0.482	0.550	0.577	0.578	0.592	0.592	0.553	0.514	0.467	0.441	0.429		(40)
(41)		taud	2.029	1.947	1.798	1.787	1.842	1.891	1.936	2.039	2.098	2.124	2.108	2.063		(41)
(42)		Ebn at Noon	733	745	728	734	739	727	724	745	754	754	728	714		(42)
(43)		Edh at Noon	136	164	206	218	208	198	188	167	151	136	125	125		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.10	2.83	3.90	4.79	5.24	4.53	4.86	5.10	4.03	3.38	2.49	1.94		(44)
(45)		RadStd	0.19	0.26	0.28	0.42	0.58	0.49	0.71	0.54	0.42	0.35	0.21	0.13		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page